Mohammad Mahdi Khadem

Medical Doctor · AI Researcher · Computational Medicine

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Professional Experience

2026 - Now **Principal Medical AI Product & Innovation Lead**, AIMedic, Tehran, Iran (remote-first)
Lead the design, development, and evaluation of the company's clinical imaging AI portfolio, spanning LLM and vision-language model products, mammography and brain-CT image analysis, and structured radiology reporting. Own the full path from research prototype to pilot-ready product, including model benchmarking, evidence documentation, and clinical workflow integration. Read more about company

2025 - Now **Medical Insurance Consultant (AI & Analytics)** , Saman Insurance Company, Tehran, Iran
Dedicated Medical Reviewer specializing in clinical documentation integrity and the integration of AI in insurance medicine. Conducts research in insurance medicine to support evidence-based policy development. Designed and implemented industrial AI agents and human-in-the-loop systems for health insurance underwriting and claim processing workflows.

2024 - Now **AI Researcher and Developer**, Minimally Invasive Research Center, Iran University of Medical Sciences
Develops AI-based clinical decision support for laparoscopic bariatric procedures using multimodal clinical datasets. Built computer vision models for real-time surgical video analysis, tool tracking, and automatic measurement of biliopancreatic limb length. Created a RAG-based clinical question-answering system grounded in the ASMBS textbook.

2022 - 2024 **General Practitioner, Outpatient Clinic**, Foolad Clinic, Tehran
Provided comprehensive primary care to geriatric patients, with an emphasis on managing chronic multi-morbidities like neuropsychiatric conditions

Education

Iran University of Medical Sciences, School of Medicine *Tehran*
M.D. (Medical Doctor) *2015 - 2022*

- Title: Evaluation of Neuroimaging Findings and Neuropsychological Test Data in Frontotemporal Dementia Patients
- Thesis Summary: Identified structural MRI correlates of cognitive impairment in frontotemporal dementia by linking routine 1.5T clinical MRI data with MMSE and MoCA sub-scores using slice-to-volume reconstruction and quantitative neuroimaging.
- Promoters: Prof. Mostafa Almasi-Dooghaee, Prof. Leila Kamalzadeh, Prof. Behnam Shariati
- Defense date: 05/12/2022
- More Detail About This Project

Publications in short

8 manuscripts: 4 published (including one meta-analysis), 1 preprint posted on Research Square (under review at *Obesity Surgery*), and 3 in preparation. A complete publication list appears at the end of this CV.

Reviewer

I served as a peer reviewer for journals in the field of Medicine, including *The American Journal of Case Reports* and *Medical Science Monitor*.

Awards and grants

2025	**Research Grant: PERSiMED — Persian Intelligent Medical Document Reader**, Saman Insurance Company	*€1,000*
2024	**Research Grant: Biliopancreatic Limb Length — From Estimation to Measurement (Pilot Study)**, Minimally Invasive Research Center	*€800*
2024	**Research Grant: Effects of Metabolic and Bariatric Surgery on Serum Uric Acid**, IUMS(IR.IUMS.REC.1403.882)	*€300*
2023	**Research Grant: Gastric Botulinum Toxin Injection and Weight Loss**, IUMS(IR.IUMS.REC.1403.884)	*€300*
2015	**Top 1% Rank Nationwide**, Iranian University Entrance Examination (Konkur)	

Teaching and supervision

Teaching

2023	**Meta-Analysis Methods for Neuroimaging Studies , Lecturer**, A hands-on workshop focused on conducting meta-analyses in neuroimaging studies, covering study selection, effect size extraction, and quantitative synthesis to draw robust, reproducible conclusions from brain imaging data.	*National Brian Center, Tehran*
2018	**1st National Health Hackathon, lead organizer**, Designed a solution, guided by MIT Health Hackathon frameworks, to bridge collaboration between clinicians and engineers in addressing real-world clinical and healthcare challenges.	*IUMS*
2017	**Advanced Python Programming for Medical Students, Lecturer**, An advanced course that trains medical students to use Python libraries such as scikit-learn and statsmodels for statistical analysis, modeling, and data-driven medical research.	*IUMS*
2017	**Basic Python Programming for Medical Students, Lecturer**, An introductory programming course designed to equip medical students with foundational Python skills for data handling and analysis.	*IUMS*

Membership

Research community engagement

2025 - now	**iAIMS (International Community for AI in MS)**, Board candidate, Coding Lead (*see the boards*). The elected board leads the community's coding activities, journal clubs, and educational sessions for AI in multiple sclerosis	*International*

Professional Membership

2022-Now	**Iran Medical Council (IRIMC)**, Membership code: 206373	*Iran*

Technical Skills

Programming and Software

Python	**Advanced**	R	**Intermediate**	MATLAB	**Intermediate**
HTML/CSS	**Intermediate**	JavaScript	**Basic**	Git	**Proficient**
Linux	**Proficient**	Docker	**Proficient**	SQL	**Proficient**

Neuroimaging

SPM	**Intermediate**	FreeSurfer	**Proficient**	ITK-SNAP	**Proficient**

Statistics & Modeling

SPSS	**Intermediate**	Study Design	**Proficient**	Mplus	**Intermediate**
Stata	**Intermediate**	Meta-analysis	**Proficient**	SEM	**Proficient**

LLMs & RAG

LangChain	**Proficient**	Ollama	**Proficient**	ChromaDB	**Proficient**
RAG Pipelines	**Proficient**				

Languages

English **Advanced** German **Basic proficiency** Persian **Native**

Publications

* indicates corresponding authorship.

Published

Arshad Ali , Shahab Shahabi , Delaram Moosavi , **SMM Khadem*** , ... , Mohammad Kermansaravi 2025. Unveiling Predictors of Hypoalbuminemia Following One Anastomosis Gastric Bypass (OAGB): A Retrospective Analysis Obesity Surgery by Springer Link **IF:3.1 ,Q1(2024)** More Detail About This Project

Delaram Moosavi , **SMM Khadem*** , ... , Azadeh Goodarzi 2025. Navigating the Complexities of Pemphigus Vulgaris: A Comprehensive Iranian Study Immunity, Inflammation and Disease published by Wiley **IF:2.7 ,Q1(2024)** More Detail About This Project

Delaram Moosavi , **SMM Khadem*** , ... , Seyed Arash Javadmoosavi 2025. Effects of Metabolic and Bariatric Surgery on Serum Uric Acid and its relation with Renal Function, and Visceral Adiposity: A Systematic Review and Meta-Analysis Obesity Surgery by Springer Link **IF:3.1 ,Q1(2024)** More Detail About This Project

SMM Khadem*, Jamil Seydi Moradi, 2025. PERSiMED: PERSian Intelligent MEDical Document Reader: A Smart NLP-based Framework for Persian Medical Documents. The 32nd National Conference and the 13th International Conference on Insurance and Development: 'Reimagining Supervision in Insurance: An Innovative Path toward Effective Regulation' More Detail About This Project

Preprint

SMM Khadem*, Delaram Moosavi, Mohammad Qasim Sorush, Mohammad Kermansaravi, Shahab Shahabi 2026. Retrieval-Augmented Clinical Question-Answering System for Bariatric Surgery Built on the ASMBS Textbook: From Good to Great Research Square, DOI: 10.21203/rs.3.rs-9979413/v1 (under review at *Obesity Surgery*) More Detail About This Project

Ongoing Project

SMM Khadem* , AmirHussein Abdolalizadeh , ApoE Genotype Related Differences in Hippocampal Shape and Subfield Structure in Cognitively Normal Adults More Detail About This Project

SMM Khadem*, ..., Mohammad Kermansaravi . 2025. Biliopancreatic Limb Length: From Estimation to Measurement: A Pilot Study More Detail About This Project

SMM Khadem* , ... , Mohammad Kermansaravi, 2025. Modeling Long-Term Weight Trajectories After Bariatric Surgery Using Latent Class Analysis and Explainable Machine Learning More Detail About This Project

References

Dr Shahab Shahabi Shahmiri (M.D., Ph.D.)
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Dr Mohammad Kermansaravi (M.D.)
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